

C PROGRAMMING :

```
Session Actions Edit View Help
C:\Users\HP\Documents> gcc -o adl adl.c
adl.c
#include <stdio.h>
int main()
{
    int Allocation[10][10], request[10][10], available[10], work[10], finish[10], n, m, i, j, k, found;
    printf("Enter Number of Processes: "); scanf("%d", &n);
    printf("Enter Number of Resource Types: "); scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            scanf("%d", &Allocation[i][j]);
    printf("\nEnter Request Matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            scanf("%d", &request[i][j]);
    printf("\nEnter Available Resources:\n");
    for (i = 0; i < m; i++)
        scanf("%d", &available[i]);
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            if (request[i][j] > available[j])
                doFound = 0;
    for (i = 0; i < n; i++)
        if (finish[i] == 0)
            if (request[i][0] > work[0])
                if (i == 0)
                    for (k = 0; k < m; k++)
                        work[k] = Allocation[i][k];
                    if (finish[i] == 1)
                        found = 1;
                    while (found == 0)
                        printf("No Deadlocked Processes\n");
                    for (j = 0; j < m; j++)
                        if (request[i][j] == 0)
                            printf("P%d ", i);
                        if (found == 1)
                            printf("No Deadlock Detected\n");
                            printf("\n");
                            return 0;
}
```

OUTPUT :

```
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C:\Users\HP\Documents> gcc -o adl adl.c
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#include <stdio.h>
int main()
{
    int Allocation[10][10], request[10][10], available[10], work[10], finish[10], n, m, i, j, k, found;
    printf("Enter Number of Processes: "); scanf("%d", &n);
    printf("Enter Number of Resource Types: "); scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            scanf("%d", &Allocation[i][j]);
    printf("\nEnter Request Matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            scanf("%d", &request[i][j]);
    printf("\nEnter Available Resources:\n");
    for (i = 0; i < m; i++)
        scanf("%d", &available[i]);
    for (i = 0; i < n; i++)
        for (j = 0; j < m; j++)
            if (request[i][j] > available[j])
                doFound = 0;
    for (i = 0; i < n; i++)
        if (finish[i] == 0)
            if (request[i][0] > work[0])
                if (i == 0)
                    for (k = 0; k < m; k++)
                        work[k] = Allocation[i][k];
                    if (finish[i] == 1)
                        found = 1;
                    while (found == 0)
                        printf("No Deadlocked Processes\n");
                    for (j = 0; j < m; j++)
                        if (request[i][j] == 0)
                            printf("P%d ", i);
                        if (found == 1)
                            printf("No Deadlock Detected\n");
                            printf("\n");
                            return 0;
}
```